

4.5.7 Momentum (HT only)

4.5.7.1 Momentum is a property of moving objects

Content	Key opportunities for skills development
Momentum is defined by the equation: $\text{momentum} = \text{mass} \times \text{velocity}$ $p = m v$ momentum, p, in kilograms metre per second, kg m/s mass, m, in kilograms, kg velocity, v, in metres per second, m/s	WS 1.2 MS 3b, c Students should be able to recall and apply this equation.

4.5.7.2 Conservation of momentum

Content	Key opportunities for skills development
In a closed system, the total momentum before an event is equal to the total momentum after the event. This is called conservation of momentum. Students should be able to use the concept of momentum as a model to: - describe and explain examples of momentum in an event, such as a collision (physics only) complete calculations involving an event, such as the collision of two objects.	AT 1, 2, 3 Investigate collisions between laboratory trolleys using light gates, data loggers or ticker timers to measure and record data.

4.5.7.3 Changes in momentum (physics only)

Content	Key opportunities for skills development
When a force acts on an object that is moving, or able to move, a change in momentum occurs.	
The equations $F = m \times a$ and $a = \frac{v-u}{t}$ combine to give the equation $F = \frac{m \Delta v}{\Delta t}$ where $m\Delta v$ = change in momentum ie force equals the rate of change of momentum.	MS 3b, c Students should be able to apply this equation which is given on the Physics Equation sheet.

Content	Key opportunities for skills development
Students should be able to explain safety features such as: air bags, seat belts, gymnasium crash mats, cycle helmets and cushioned surfaces for playgrounds with reference to the concept of rate of change of momentum.	WS 1.2, 4
Students should be able to apply equations relating force, mass, velocity and acceleration to explain how the changes involved are inter-related.	MS 3b, 3c, 3d